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TRANSPORT PHENOMENA I, CHE 311

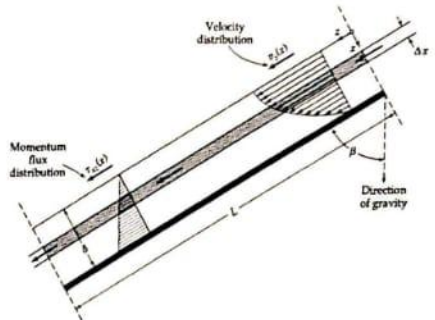
QUIZ 1

CLOSED BOOK, TIME: 15 MINUTES

The velocity profile of a liquid film flowing down an inclined flat plate is given by:

$$v_z = \frac{\rho g \cos \beta \delta^2}{2\mu} \left(1 - \frac{x^2}{\delta^2}\right)$$

- 1- Write down an equation for the z-component of the mass flux as a function of x.
- 2- Calculate the viscous tension on the gas-liquid and the solid-liquid interfaces.



All your answers must be in terms of the following constant parameters.

μ and ρ : viscosity and density of the liquid

δ : film thickness

β : angle of the flat plate relative to the direction of gravity

g : gravitational constant

Newton's law of viscosity: $|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$

BSL textbook notation: $\tau = -\mu(\nabla v + (\nabla v)^T)$

1.) z comp. mass flux - $\rho \cdot v_z$ ✓ 50

$$f(x) = \frac{\rho^2 g \cos \beta \delta^2}{2\mu} \left(1 - \frac{x^2}{\delta^2}\right)$$

$$\frac{\partial v_z}{\partial x}$$

$$\frac{50}{100}$$

$$|\tau| = -\mu \left(\begin{vmatrix} 0 & 0 & \frac{\partial v_z}{\partial x} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial x} & 0 & 0 \end{vmatrix} \right) \Rightarrow 0$$

2.) $\rho \cdot v_z = \left\{ \begin{matrix} \text{molecular} \\ \text{convection} \end{matrix} \right\} + \left\{ \right\} - \left\{ \right\} - \rho g_x$

$$v_z$$